Modulus Functions & the Factor/Remainder Theorem

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Consider the function $f(x) = \frac{ax^2 + bx + c}{dx + e}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 2$ and a horizontal asymptote at $y = 3$. The graph also passes through the points $(0, 0)$ and $(1, -3)$. Find the value of $\frac{a}{d}$.

Final Answer: 3. The vertical asymptote at $x = 2$ implies that the denominator $dx + e$ is zero when $x = 2$, so $2d + e = 0$, which means $e = -2d$. Thus, the denominator can be written as $d(x - 2)$. The function becomes $f(x) = \frac{ax^2 + bx + c}{d(x - 2)}$. The horizontal asymptote at $y = 3$ implies that the limit of $f(x)$ as $x \to \pm\infty$ is 3. For a rational function, this limit is the ratio of the leading coefficients if the degree of the numerator is equal to the degree of the denominator.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Consider the function $f(x) = \frac{ax^2 + bx + c}{x - d}$, where $a, b, c, d \in \mathbb{R}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 3$ and a horizontal asymptote at $y = 2$. The function passes through the point $(1, -8)$. If $f'(x)$ has a critical point at $x = 5$, find the value of d.

The vertical asymptote at $x = 3$ implies that the denominator is zero at $x = 3$, so $d = 3$. The horizontal asymptote $y = 2$ for a rational function where the degree of the numerator is equal to the degree of the denominator is the ratio of the leading coefficients. Therefore, $\frac{a}{1} = 2$, which means $a = 2$. Now we have $f(x) = \frac{2x^2 + bx + c}{x - 3}$. The function passes through $(1, -8)$: $-8 = \frac{2(1)^2 + b(1) + c}{1 - 3} = \frac{2 + b + c}{-2}$ $16 = 2 + b + c \Rightarrow b + c = 14$. Next, we find the.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

[Vectors & 3D Geometry · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

4 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

[Statistics & Probability (HL) · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

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5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Calculus (HL) · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Functions: Modulus & Inverse · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Given $f(x) = 2x - 3$ for $x \in \mathbb{R}$, sketch $y = |f(x)|$ and solve $|2x - 3| = 5$.

Sketch: V-shape with vertex at $(1.5, 0)$. Solve: $2x - 3 = 5 \Rightarrow x = 4$ or $2x - 3 = -5 \Rightarrow x = -1$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 3 · Advanced For the position vector $\mathbf{r}(t) = \langle t, t^2, \frac{2}{3}t^3 \rangle$, find the tangential component of acceleration a_T and the normal component a_N at $t = 1$.

Compute derivatives: $\mathbf{r}'(t) = \langle 1, 2t, 2t^2 \rangle$ $\mathbf{r}''(t) = \langle 0, 2, 4t \rangle$ At $t = 1$:
$\mathbf{r}'(1) = \langle 1, 2, 2 \rangle \Rightarrow |\mathbf{r}'(1)| = \sqrt{1 + 4 + 4} = 3$ $\mathbf{r}''(1) = \langle 0, 2, 4 \rangle$ Tangential acceleration a_T:
$a_T = \frac{\mathbf{r}' \cdot \mathbf{r}''}{|\mathbf{r}'|} = \frac{(1)(0) + (2)(2) + (2)(4)}{3} = \frac{0 + 4 + 8}{3} = \frac{12}{3} = 4$ Normal acceleration a_N: Compute cross product at $t = 1$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 3 · AdvancedWater flows into an initially empty tank at a rate $R(t) = 120 - 4t$ liters/hour and leaks out at a rate $L(t) = 3t$ liters/hour for $0 \le t \le 24$ hours. Find the maximum volume of water in the tank and the time at which the tank becomes completely empty.

Step 1 · Volume Accumulation Function: Net rate of change is
$V'(t) = R(t) - L(t) = (120 - 4t) - 3t = 120 - 7t$. Since the tank is initially empty, $V(0) = 0$:
$V(t) = \int_0^t (120 - 7u)\, du = 120t - \frac{7}{2}t^2$. Step 2 · Maximum Volume: Maximum occurs when
$V'(t) = 0 \Rightarrow 120 - 7t = 0 \Rightarrow t = \frac{120}{7} \approx 17.14$ hours. Since x for $t > 0$:
$120t - \frac{7}{2}t^2 = 0 \Rightarrow t(120 - \frac{7}{2}t) = 0 \Rightarrow t = \frac{240}{7} \approx 34.29$ hours. Since t.

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10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Level 1 · FoundationsVerify that the Mean Value Theorem applies to $f(x) = x + \frac{1}{x}$ on $[1, 4]$, and find all values of c that satisfy the theorem.

Step 1 · Check Hypotheses: $f(x)$ is continuous on $[1, 4]$ and differentiable on $(1, 4)$ since the only discontinuity $x = 0$ lies outside $[1, 4]$. The hypotheses of MVT hold. Step 2 · Compute Average Rate of Change: $\frac{f(4) - f(1)}{4 - 1} = \frac{(4 + 1/4) - (1 + 1)}{3} = \frac{17/4 - 2}{3} = \frac{9/4}{3} = \frac{3}{4}$. Step 3 · Solve $f'(c) = \frac{3}{4}$.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Level 3 · Advanced Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x^3, y^3, z^3 \rangle$ across the boundary sphere $S : x^2 + y^2 + z^2 = 1$.

Compute the divergence: $abla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x^3) + \frac{\partial}{\partial y}(y^3) + \frac{\partial}{\partial z}(z^3) = 3x^2 + 3y^2 + 3z^2 = 3(x^2 + y^2 + z^2)$ Apply the Divergence Theorem: $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E 3(x^2 + y^2 + z^2)\, dV$ Convert to spherical coordinates: $x^2 + y^2 + z^2 = \rho^2$, $dV = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Determine the slant asymptote of $g(x) = \frac{2x^3 - 3x^2 + 5}{x^2 + 1}$ and find the coordinates of any point where the curve intersects this asymptote.

Since $\deg(\text{num}) = 3$ and $\deg(\text{denom}) = 2$ with $3 - 2 = 1$, a slant asymptote exists. Perform polynomial long division: $(2x^3 - 3x^2 + 5) \div (x^2 + 1) = (2x - 3) + \frac{-2x + 8}{x^2 + 1}$. The linear quotient gives the slant asymptote: $y = 2x - 3$. To find where the graph intersects this asymptote, set the remainder equal to zero: $\frac{-2x + 8}{x^2 + 1} = 0 \Rightarrow -2x + 8 = 0 \Rightarrow x = 4$. Substitute $x = 4$ into the asymptote equation: $y = 2(4) - 3 = 5$. The curve intersects.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Level 2 · Medium Find the area of the triangle with vertices $A(1, 1, 0)$, $B(3, 0, 2)$, and $C(0, 2, 4)$.

Find vectors $\vec{AB}$ and $\vec{AC}$: $\vec{AB} = \langle 2, -1, 2 \rangle$ $\vec{AC} = \langle -1, 1, 4 \rangle$ Compute the cross product: $\vec{AB} \times \vec{AC} = |\mathbf{i}, \mathbf{j}, \mathbf{k}; 2, -1, 2; -1, 1, 4| = \mathbf{i}(-4 - 2) - \mathbf{j}(8 - (-2)) + \mathbf{k}(2 - 1) = \langle -6, -10, 1 \rangle$ Magnitude of cross product: $|\vec{AB} \times \vec{AC}| = \sqrt{(-6)^2 + (-10)^2 + 1^2} = \sqrt{36 + 100 + 1} = \sqrt{137}$ The triangle area is half the parallelogram area.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Level 3 · Advanced Evaluate the double integral $\iint_D e^{x^2 + y^2}\, dA$ where D is the semi-annular region $\{(x, y) : 1 \le x^2 + y^2 \le 4, y \ge 0\}$.

Convert the semi-annulus to polar coordinates: Radius bounds: $1 \le r^2 \le 4 \Rightarrow 1 \le r \le 2$. Angle bounds (since $y \ge 0$, upper half-plane): $0 \le \theta \le \pi$. Area element: $dA = r\, dr\, d\theta$. Set up the integral: $\iint_D e^{x^2 + y^2}\, dA = \int_0^{\pi} \int_1^2 e^{r^2}\, r\, dr\, d\theta = \left(\int_0^{\pi} d\theta\right)\left(\int_1^2 r e^{r^2}\, dr\right)$ Substitute $u = r^2 \Rightarrow du = 2r\, dr$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity**

Find all vertical and horizontal asymptotes, the exact coordinates of any holes, and all intercepts of the rational function $R(x) = \frac{2x^2 + 5x - 3}{x^2 + 4x + 3}$.

Factor numerator and denominator completely: $R(x) = \frac{(2x-1)(x+3)}{(x+1)(x+3)}$. The factor $(x+3)$ cancels for $xeq -3$. Evaluate the simplified form $g(x) = \frac{2x-1}{x+1}$ at $x = -3$: $g(-3) = \frac{2(-3)-1}{-3+1} = \frac{-7}{-2} = 3.5$. Key features:Hole: $(-3, 3.5)$Vertical asymptote: $x + 1 = 0 \Rightarrow x = -1$Horizontal asymptote: Equal degrees (2 and 2) $\Rightarrow y = 2/1 = 2$$x$-intercept: Numerator root $2x - 1 = 0 \Rightarrow (0.5, 0)$$y$-intercept.

16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 1.04 Partial Fractions**

Find the 6th term in the expansion of $(3x - 2y^2)^8$ in fully simplified form.

In the binomial theorem expansion $\sum_{k=0}^{n} \binom{n}{k} A^{n-k} B^k$, the 1st term has $k = 0$, so the 6th term has $k = 5$. Identify terms: $n = 8$, $A = 3x$, $B = -2y^2$: $T_6 = \binom{8}{5}(3x)^{8-5}(-2y^2)^5 = \binom{8}{3}(3x)^3(-2y^2)^5$. Compute each factor:$\binom{8}{3} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56(3x)^3 = 27x^3(-2y^2)^5 = -32y^{10}$Multiply the coefficients:

$T_6 = 56 \times 27 \times (-32)x^3y^{10} = 56 \times (-864)x^3y^{10} = -48384x^3y^{10}$.

17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)**

The function f is defined by $f(x) = x^2 - 6x + 13$ for $x \geq 3$. (a) State the range of f. (b) Find an expression for $f^{-1}(x)$ and state the domain and range of f^{-1}.

(a) Complete the square: $f(x) = (x-3)^2 - 9 + 13 = (x-3)^2 + 4$. Since $x \geq 3$, $(x-3)^2 \geq 0$, so the minimum value is $f(3) = 4$. The range of f is $f(x) \geq 4$. (b) Set $y = (x-3)^2 + 4$. Rearrange for x: $y - 4 = (x-3)^2 \Rightarrow x - 3 = +\sqrt{y-4}$ (since $x \geq 3$) $x = 3 + \sqrt{y-4}$ Therefore, $f^{-1}(x) = 3 + \sqrt{x-4}$. Domain of f^{-1} is $x \geq 4$ (the range of f), and range of f^{-1} is $f^{-1}(x) \geq 3$ (the domain of f).

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 3.01 3D Vectors & the Cross Product**

Functions g and h are defined by $g(x) = 2x - 1$ for $x \in \mathbb{R}$, and $h(x) = \frac{6}{x-2}$ for $xeq 2$. (a) Find an expression for $hg(x)$ in simplest form. (b) Solve the equation $hg(x) = g(x)$.

(a) Substitute $g(x)$ into h: $hg(x) = h(2x-1) = \frac{6}{(2x-1)-2} = \frac{6}{2x-3}$ $\left(xeq \frac{3}{2}\right)$ (b) Set $hg(x) = g(x)$: $\frac{6}{2x-3} = 2x - 1 \Rightarrow 6 = (2x-1)(2x-3)$ Expand the right-hand side: $6 = 4x^2 - 8x + 3 \Rightarrow 4x^2 - 8x - 3 = 0$ Factor: $(2x-3)(2x+1) = 0$. Since $xeq \frac{3}{2}$ for $hg(x)$ to be defined, the value $x = \frac{3}{2}$ is excluded. Therefore, the unique solution is $2x + 1 = 0 \Rightarrow x = -\frac{1}{2}$.

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19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Level 2 · Medium Evaluate the iterated integral by reversing the order of integration: $\int_0^{\pi}\int_y^{\pi} \frac{\sin x}{x}\,dx\,dy$

The domain is $0 \le y \le \pi$ and $y \le x \le \pi$. This is the triangular region bounded by $y = 0$, $x = \pi$, and $y = x$. Reversing order to Type I: x ranges from 0 to π, while y ranges from 0 to x:

$\int_0^{\pi}\int_0^{x} \frac{\sin x}{x}\,dy\,dx = \int_0^{\pi}\left[y\frac{\sin x}{x}\right]_{y=0}^{y=x} dx = \int_0^{\pi} x\frac{\sin x}{x}\,dx = \int_0^{\pi}\sin x\,dx = [-\cos x]_0^{\pi} = -\cos(\pi) - (-\cos 0) = -(-1) - (-1) = 1 + 1 = 2$

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Level 2 · Medium Use Green's Theorem to evaluate $\oint_C (3y\,dx + 5x\,dy)$ where C is the counterclockwise circle $x^2 + y^2 = 4$.

Identify components: $P = 3y$ and $Q = 5x$. Compute partial derivatives: $\frac{\partial Q}{\partial x} = 5$ and $\frac{\partial P}{\partial y} = 3$. Apply Green's Theorem: $\oint_C (3y\,dx + 5x\,dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA = \iint_R (5 - 3)\,dA = 2\iint_R dA.$ The region R is a disk of radius $r = 2$, so its area is $\pi(2^2) = 4\pi$. Therefore, $2\iint_R dA = 2(4\pi) = 8\pi$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Why can $|x - 7| = -3$ have no real solutions?

a. The absolute value of any real expression is non-negative ($|u| \ge 0$), so it cannot equal a negative number ✓

b. Because x must be greater than 7

c. Because -3 is not a prime number

d. The equation has complex solutions only

By definition, $|u| \ge 0$ for all $u \in \mathbb{R}$. A non-negative quantity can never equal -3.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve the inequality $|2x - 3| < 7$.

a. $-2 < x < 5$ ✓

b. $x < 5$

c. $x < -2$ or $x > 5$

d. $-5 < x < 5$

$|2x - 3| < 7 \Leftrightarrow -7 < 2x - 3 < 7$. Add 3: $-4 < 2x < 10$. Divide by 2: $-2 < x < 5$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which interval notation represents the solution to $|x - a| \leq \varepsilon$ where $\varepsilon > 0$?

a. $[a - \varepsilon, a + \varepsilon]$ ✓

b. $(a - \varepsilon, a + \varepsilon)$

c. $[-\varepsilon, \varepsilon]$

d. $[a, a + \varepsilon]$

$|x - a| \leq \varepsilon \Leftrightarrow -\varepsilon \leq x - a \leq \varepsilon \Leftrightarrow a - \varepsilon \leq x \leq a + \varepsilon$, giving $[a - \varepsilon, a + \varepsilon]$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

When $P(x) = 2x^3 - 3x^2 + ax - 5$ is divided by $(x - 2)$, the remainder is 11. Find a.

a. 6 ✓

b. 4

c. -6

d. 8

By the Remainder Theorem, $P(2) = 11$. $P(2) = 2(8) - 3(4) + 2a - 5 = 16 - 12 + 2a - 5 = 2a - 1$. Setting $2a - 1 = 11 \Rightarrow 2a = 12 \Rightarrow a = 6$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $(x + 3)$ is a factor of $Q(x) = x^3 + kx^2 - 7x + 6$, find k.

a. 2 ✓

b. -2

c. 3

d. -1

By the Factor Theorem, $Q(-3) = 0$. $Q(-3) = (-3)^3 + k(-3)^2 - 7(-3) + 6 = -27 + 9k + 21 + 6 = 9k$. Setting $9k = 0 \Rightarrow k = 0$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A polynomial $f(x)$ leaves remainder 3 when divided by $(x - 1)$ and remainder 7 when divided by $(x - 3)$. What is the linear remainder when $f(x)$ is divided by $(x - 1)(x - 3)$?

a. $2x + 1$ ✓

b. $2x - 1$

c. $x + 2$

d. $4x - 1$

Let $R(x) = ax + b$. $R(1) = a + b = 3$ and $R(3) = 3a + b = 7$. Subtracting: $2a = 4 \Rightarrow a = 2$. Then $b = 3 - 2 = 1$. The remainder is $2x + 1$.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

How is the graph of $y = |f(x)|$ obtained from the graph of $y = f(x)$?

a. Any part of the graph lying below the x-axis is reflected vertically across the x-axis into the upper half-plane ✓

b. The left half of the graph is discarded and replaced by reflection of the right half

c. The graph is shifted 1 unit upward

d. The graph is reflected across the y-axis

Taking $|f(x)|$ preserves positive y-values and takes the opposite of negative y-values, reflecting sections below the x-axis upward across the x-axis.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

How is the graph of $y = f(|x|)$ obtained from $y = f(x)$?

a. Discard the portion for $x < 0$, and reflect the portion for $x \geq 0$ across the y-axis to form an even function ✓

b. Reflect the entire graph across the x-axis

c. Reflect the graph across $y = x$

d. Shift the graph to the right by $|x|$

Because $|-x| = |x|$, $f(|x|)$ is an even function: the right half ($x \geq 0$) is reflected across the y-axis, discarding the original $x < 0$ part.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the vertex (sharp turn) of the function $g(x) = 3|x - 4| - 5$?

a. $(4, -5)$ ✓

b. $(-4, -5)$

c. $(4, 5)$

d. $(0, -5)$

The absolute value term $|x - 4|$ reaches its minimum of 0 at $x = 4$. At $x = 4$, $g(4) = 0 - 5 = -5$, giving vertex $(4, -5)$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Solve $|x + 1| = |-3x - 1|$. Give the largest solution.

a. -2

b. 4

c. 3

d. **0** ✓

Working through this functions step: Case 1: $x + 1 = -3x - 1$. Case 2: $x + 1 = -(-3x - 1)$. Largest solution: $x = 0$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Factor $P(x) = x^3 - x^2 - 4x + 4$ completely. Find the largest root.

a. **2** ✓

b. 1

c. 4

d. 3

Working through this additional-topics step: $P(x) = (x - 1)(x - 2)(x + 2) \Rightarrow$ roots $= 1, 2, -2$; largest $= 2$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find the remainder when $P(x) = 2x^2 + 3x - 4$ is divided by $(x - 4)$.

a. **40** ✓

b. 32

c. 29

d. 47

Working through this additional-topics step: By the Remainder Theorem, remainder $= P(4) = 2(4)^2 + 12 - 4 = 40$

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve $|x - 6| = 2x - 9$, given the solution is positive.

a. 6

b. 7

c. **5** ✓

d. 4

Working through this functions step: Testing $x = 5 : |5 - 6| = 1,$ and $2(5) - 9 = 1.$ Verified.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Is $f(x) = 2x^3$ odd, even, or neither?

a. Odd ✓

b. Neither

c. Both odd and even

d. Even

Comparing $f(-x)$ with $f(x)$ and $-f(x)$ shows the function is odd.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Given that $(x - 2)$ is a factor of $P(x) = 3x^2 - 6 + kx$, find k.

a. -7

b. -4

c. **-3** ✓

d. -2

Working through this additional-topics step: $P(2) = 0 \Rightarrow 3(2)^2 + k(2) - 6 = 0 \Rightarrow k = -3$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Evaluate $\tan^{-1}(1)$ in degrees.

a. 67°

b. 22°

c. 69°

d. 45° ✓

Working through this functions step: $\tan^{-1}(1) = 45^\circ$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Given that f is an even function and $f(9) = 10$, find $f(-9)$.

a. 4

b. **10** ✓

c. 13

d. 14

Even : $f(-x) = f(x) \Rightarrow f(-9) = f(9) = 10$

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 3.01 3D Vectors & the Cross Product**

Given that f is an odd function and $f(6) = -5$, find $f(-6)$.

a. 8

b. 1

c. 2

d. **5 ✓**

Odd : $f(-x) = -f(x) \Rightarrow f(-6) = -(-5) = 5$

39 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 3.02 Lines and Planes in Three Dimensions**

If $f(x) = \frac{4}{x}$ (a self-inverse function), find the positive value of x for which $f(x) = x$ (a fixed point).

a. **2 ✓**

b. 1

c. 3

d. 4

$\frac{4}{x} = x \Rightarrow x^2 = 4 \Rightarrow x = 2$

40 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)**

Using the Remainder Theorem, find the remainder when $f(x) = 2x^2 - 1x + 6$ is divided by $x + 3$.

a. -27

b. 28

c. **27 ✓**

d. 26

By the Remainder Theorem, the remainder equals $f(-3) = 2(-3)^2 - 1(-3) + 6 = 27$.

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Consider the function $f(x) = \frac{ax^2 + bx + c}{x - d}$, where a, b, c, and d are constants. The graph of $f(x)$ has a vertical asymptote at $x = 3$ and an oblique asymptote $y = 2x + 1$. Given that $f(1) = -4$ and $f(2) = -6$, find the value of a.

a. -2

b. 2 ✓

c. -4

d. 4

The vertical asymptote occurs where the denominator is zero, so $x - d = 0$. Given the vertical asymptote is at $x = 3$, we have $d = 3$. Thus, $f(x) = \frac{ax^2 + bx + c}{x - 3}$. For the oblique asymptote, we perform polynomial long division or consider the limit as $x \to \infty$. The oblique asymptote is $y = 2x + 1$. This implies that as $x \to \infty$, $\frac{ax^2 + bx + c}{x - 3} \approx 2x + 1$. This can be seen by dividing the numerator by the denominator: $ax + b + \frac{c'}{x - 3}$. For the oblique asymptot.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.02 The Poisson Distribution

Consider the tetrahedron OABC with vertices O(0, 0, 0), A(2, 1, 3), B(1, 2, -1), and C(k, k+1, k-1). Given that the volume of the tetrahedron is 7 cubic units, find the possible values of k.

a. k = 1 or k = 4

b. **k = 2 or k = 3** ✓

c. k = 3 or k = 4

d. k = 4 or k = 5

The volume of a tetrahedron with vertices O(0,0,0), A, B, C is given by the formula $V = \frac{1}{6}|\det(\vec{OA}, \vec{OB}, \vec{OC})|$. Given the vertices O(0,0,0), A(1,0,0), B(0,1,0), and C(k, k, k^2 - 5k + 48), the vectors from the origin are: $\vec{OA} = (1\ 0\ 0)$, $\vec{OB} = (0\ 1\ 0)$, and $\vec{OC} = (k\ k\ k^2 - 5k + 48)$. The determinant of these vectors is: $\det(\vec{OA}, \vec{OB}, \vec{OC}) = |1, 0, k\ 0, 1, k\ 0, 0, k^2 - 5k + 48|$ Since this i.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Further Integration Techniques

A school offers two optional subjects: Art and Music. The number of students enrolled in each subject and the number of students enrolled in both is shown in the Venn diagram below. The total number of students in the school is 120. Let A be the set of students enrolled in Art and M be the set of students enrolled in Music. Given that $n(A \cup M) = 90$, find the number of students enrolled in neither Art nor Music.

a. 20

b. **30** ✓

c. 40

d. 10

The Venn diagram shows that $n(A \cap M) = 20$, $n(A \setminus M) = 3x$, and $n(M \setminus A) = 4x$. The total number of students in the school is 120. We are given that $n(A \cup M) = 90$. The formula for the union of two sets is $n(A \cup M) = n(A \setminus M) + n(M \setminus A) + n(A \cap M)$. Substituting the given values: $90 = 3x + 4x + 20$. Subtracting 20 from both sides: $70 = 7x$. Dividing by 7: $x = 10$. Now we can find the number of students in Art only and Music only.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.02 Differential Equations

The following bar chart shows the number of students who chose each of four different subjects in a school. Let N be the total number of students surveyed. Find the value of N.

a. 90

b. **100** ✓

c. 96

d. 104

The bar chart shows the number of students for each subject: Subject 1 has 25 students, Subject 2 has 28 students, Subject 3 has 22 students, and Subject 4 has 25 students. The total number of students surveyed, N, is the sum of the students who chose each subject. Therefore, $N = 25 + 28 + 22 + 25 = 100$. Choice A is incorrect as it is the sum of only the first three subjects and the last subject if it were 20. Choice C is incorrect as it omits some students or miscounts the values. Choice D is incorrect as it a.

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

Consider the points A(2, 3) and B(8, 7) in the Cartesian plane. The vector $\vec{AB}$ can be represented as **v**. Find the components of **v**.

a. **(6, 4)** ✓

b. (10, 10)

c. (-6, -4)

d. (4, 6)

To find the vector $\vec{AB}$, we subtract the coordinates of the initial point A from the coordinates of the terminal point B. Let A = (x■, y■) = (2, 3) and B = (x■, y■) = (8, 7). The vector $\vec{AB}$ is given by $\mathbf{v} = (x_2 - x_1, y_2 - y_1)$. Substituting the given coordinates: $\mathbf{v} = (8 - 2, 7 - 3) = (6, 4)$. Therefore, the components of the vector $\vec{AB}$ are (6, 4). Common errors include subtracting in the reverse order (which would give (-6, -4)) or adding the coordinates (which would give (10.

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